John Leiga

johnleiga9@gmail.com • 209.822.9104 • LinkedIn • Portfolio
U.S. Citizen • Eligible to work on export-controlled programs

Education

B.S. Mechanical Engineering 2017 – 2021
University of California, Merced

Experience

Nestle Purina PetCare Company Bloomfield, MO — Ft. Dodge, IA — St. Louis, MO
Design Engineer Feb 2025 – Present

- Designed extrusion dies, production tooling, and equipment components in Autodesk Inventor based on factory needs, production feedback, and manufacturing constraints.
- Owned unit operation specifications that defined approved equipment, operating standards, and process requirements for factory systems.
- Translated factory feedback into equipment requirements, tooling updates, operating standards, and design changes to improve reliability and process consistency.
- Validated process equipment during factory startup through installation readiness checks, startup support, production ramp troubleshooting, and operational handoff.

Management Development Associate - Engineering May 2023 – Feb 2025

- Optimized a 14-furnace perlite expansion process by tuning control loops in sequence, debugging process instability, and adding density-based feedback to improve product consistency.
- Diagnosed a recurring perlite process bottleneck by tracing product backup and shutdowns to an undersized rotary airlock in the main product stream, creating the basis for equipment correction and process stabilization.
- Structured perlite DOE trials using variable screening, interaction testing, high-resolution data capture, and Minitab/Excel analysis to identify run parameters for different ore types.
- Developed and deployed a zero-waste automated startup sequence, saving $200K+ annually across Missouri and Virginia facilities.

Tavis Corporation Mariposa, CA
Project Engineer Sep 2021 – May 2023

- Designed aerospace pressure transducer components and assemblies in SolidWorks, applying stress analysis, engineering drawings, and assembly drawings to support prototype build, testing, and manufacturing documentation.
- Applied first-principles mechanical analysis to design and prototype-validate a differential pressure transducer architecture, reducing mechanical component cost by approximately 50% and total hardware cost by approximately 30%.
- Interpreted long-form aerospace specifications to define pressure range, media compatibility, environmental, test, documentation, and qualification requirements for custom hardware.
- Developed qualification test procedures to validate pressure transducer survival and operation under thermal, vibration, shock, and overpressure requirements.
- Led internal and customer-facing design reviews, aligning requirements, design tradeoffs, test readiness, qualification data, compliance documentation, open actions, and technical deliverables.
- Managed aerospace hardware programs through qualification and manufacturing integration, owning requirements, schedules, customer communication, and technical deliverables.
- Led failure analysis, documentation updates, and component obsolescence reviews to support production continuity and customer program execution.

Skills

Mechanical Design: SolidWorks, Autodesk Inventor, precision mechanical design, aerospace pressure transducer hardware, electromechanical sensor assemblies, DFM, engineering drawings, assembly drawings

Requirements and Verification: requirements interpretation, aerospace specification review, qualification planning, test procedures, compliance documentation, technical deliverables, customer-facing design reviews

Manufacturing and Product Support: production support, manufacturing documentation, production tooling, equipment components, startup validation, factory feedback, root cause analysis, failure analysis

Tools: SolidWorks, Autodesk Inventor, Microsoft Excel, Minitab, Python, MATLAB, Git

Independent Projects

Python ML Research System

- Built a modular Python research platform for equities and options strategy development, using artifact-based pipeline stages for feature engineering, labeling, LightGBM model training, backtesting, model explainability, and live inference.